The role of tongue root anteriority in the Dolgan vowel system

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The **tongue root** plays unexpected roles in the **Dolgan** vowel system

Dolgan

- Lena Turkic language, spoken on Taymyr peninsula (Arctic Siberia)
- Endangered, ~5k speakers, loss of intergenerational transmission

Phonological evidence: alternation of /G/

/G/: surfaces as velar or uvular depending on preceding vowel

Kyrgyz (&c.)

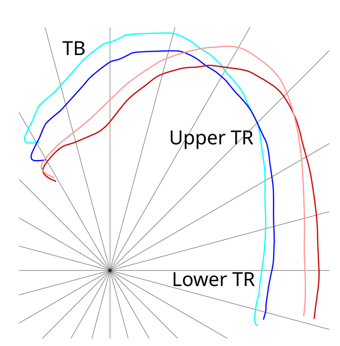
	front	back
+high	[ini-ɣe] 'brother-DAT'	[aːɾɯ-ʁa] 'bee-DAT'
−high	[ene-ɣe] 'mother-DAT'	[aɾaː-ʁa] 'saw-DAT'

- uvular G / V[+back] _, velar G / V[−back] _

↳ V backness associated with TR backness

- coproduced: TB backness & TR backness

(Washington, 2016, 2019)



Sakha

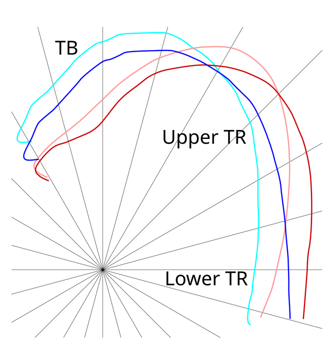
	front	back
+high	[itiː-gɛ] 'heat-DAT'	[aɾɯː-ga] 'century-DAT'
−high	[yjɛ-ʁɛ] 'century-DAT'	[taba-ʁa] 'reindeer-DAT'

- uvular G / V[−high] _, velar G / V[+high] _

↳ V height associated with TR backness

- coproduced: TB height & TR backness

(Vaux, 2009)



Dolgan

	front	back
+high	[itiː-gɛ] 'heat-DAT'	[aɾɯː-ga] 'century-DAT'
−high	[yjɛ-gɛ] 'time-DAT'	[taba-ʁa] 'reindeer-DAT'

- uvular G / V[−hi,+bk] _, velar G / elsewhere

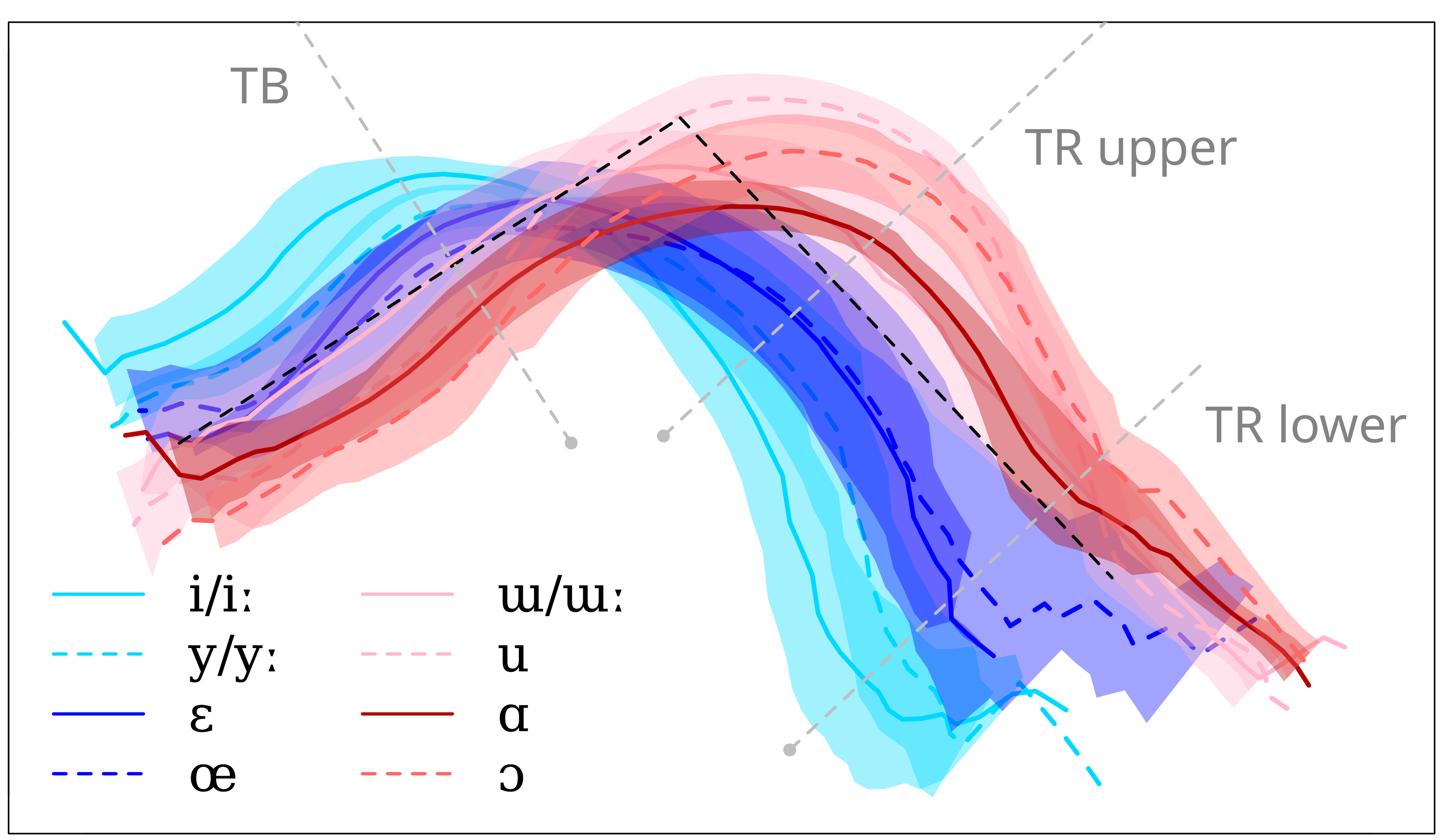
(Убрятова, 1985; Däbritz, 2022)

↳ V height & backness associated w. TR backness

- coproduced: TB height & TB & TR backness!

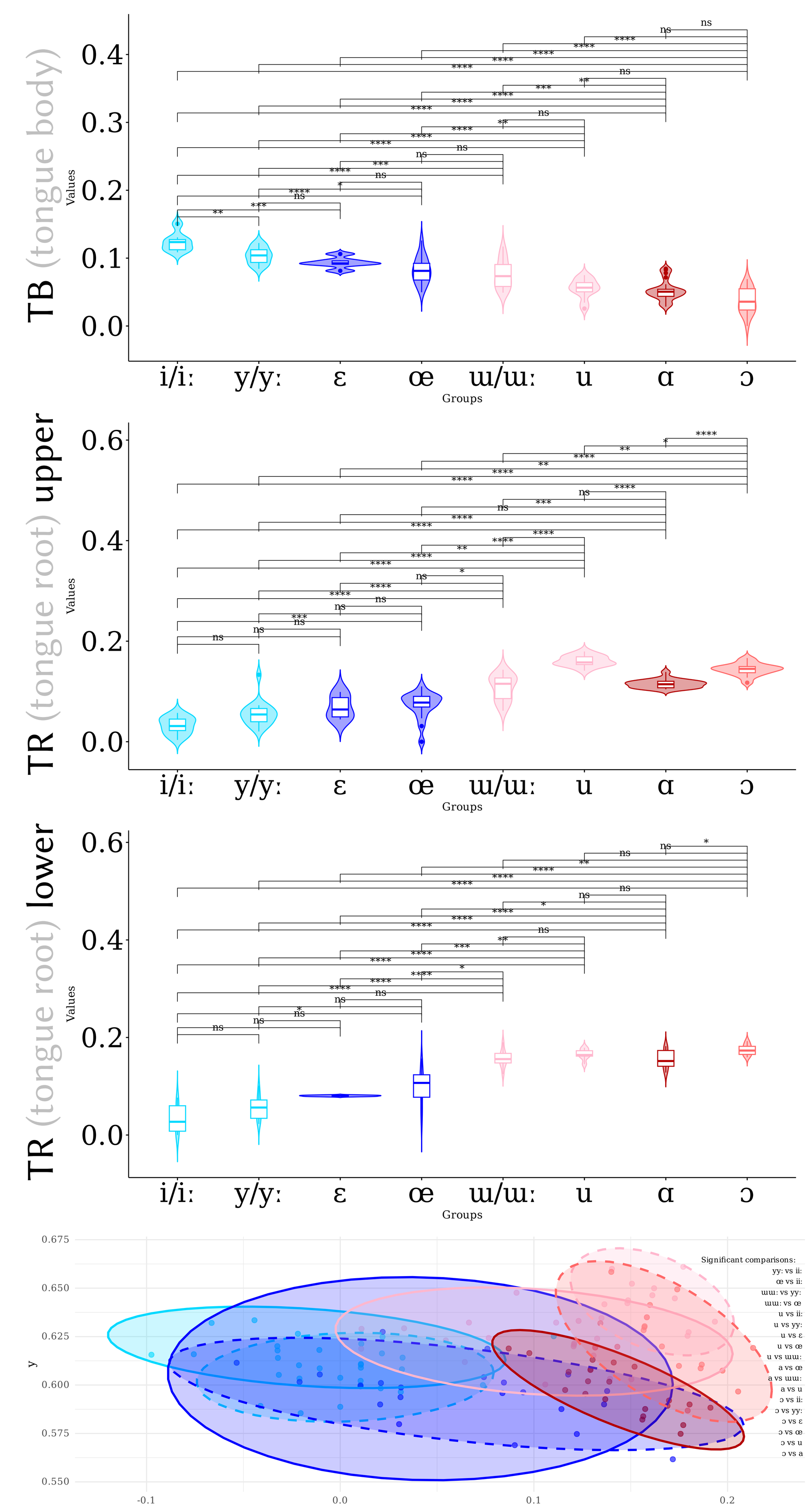
?

Results



← tongue tip left

Significance (paired t-tests)



	pair	TB middle	TB PMC	TR upper	TR lower
height	i-ɛ	**0.00052**	1.00000	0.28013	0.09608
	y-œ	0.02035	1.00000	1.00000	0.19771
	ɯ-a	**0.00555**	**0.00049**	1.00000	1.00000
	u-ɔ	0.16815	0.01150	0.02140	1.00000
backness	i-ɯ	**1.6e-07**	0.20473	**3.5e-08**	**2.4e-05**
	y-u	**1.3e-07**	**4.3e-13**	**3.8e-13**	**1.2e-07**
	ɛ-a	**5.0e-06**	1.00000	0.11347	**1.8e-09**
	œ-ɔ	**2.5e-06**	**4.2e-10**	**2.7e-08**	**0.00252**
rounding	i-y	**0.00732**	**0.00584**	0.12860	1.00000
	ɛ-œ	1.00000	1.00000	1.00000	1.00000
	ɔ-a	0.27664	**1.3e-06**	**4.3e-07**	0.03711
	u-ɯ	0.07095	**1.4e-06**	**1.4e-05**	1.00000

p values: ns ≥ 0.05 > * ≥ 0.01 > ** ≥ 0.001 > *** ≥ 0.0001 ≥ ****

Methodology

Recorded L1 speaker of Lower Xataŋga Dolgan (author 4)

- Wordlist task, carrier phrase *Зоя __ диир* 'Zoya says __' (in custom orthography)
- Articulate Instruments' Micro ultrasound system, AAA software (Wrench, 2017)

Annotated midsagittal tongue surface using UltraTrace (Murphy et al., 2020)

- frame closest to midpoint of V in 1st syllable of each word

Plotted and **analysed** using ultrapolaRplot (Outkin & Washington, 2024)

- Perpendicular rays intersect various areas of the tongue (mid TB, upper TR, etc.)
- Pairwise t-test at intersections for significance (adjusted for Type I error)
- Pairwise t-test compares maximum curvature (PMC) groups by calculating differences in distance to origin [future work: point on best fit line of curvatures]

Main findings

- V height implemented via TB height (expected)
- V height not implemented via TR backness (unexpected)
- V backness implemented via TB height (unexpected?)
- V backness implemented via TR backness (expected)
- V roundedness implemented via TB and TR (unexpected)

Impressionistically:

- 5 levels of TR backness: i > y > ɛ/œ > a/ɯ > ɔ/u

Quantitative methodology and impressionistic methodology not perfect match

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https://github.com/SwatPhonLab/ultrapolaRplot

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